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Reg. No: _____
Name: _____

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION March 2026
Fourth SEMESTER
Discipline Specific Elective (DSE) - MG4DSECHEA00 -
CHARACTERIZATION OF NANOMATERIALS
(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand (U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap). Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

PART A (Short Answer)

Answer any 5 questions. Each question carries 4 marks. (5 x 4 = 20 Marks)

1. Differentiate between KLL and LMM emission in AES. *[U] / [CO3]*
   In Auger Electron Spectroscopy (AES), transitions are named based on the three electron shells involved. The first letter indicates the shell where the initial hole is created, the second indicates the shell of the electron that fills the hole, and the third indicates the shell from which the Auger electron is ejected. In KLL emission the incident beam creates a vacancy in the innermost **K-shell** and the electron from the **L-shell** drops down to fill this vacancy. The energy released is then transferred to a second electron in the **L-shell**, which is ejected. While in LMM emission the initial vacancy is created in the **L-shell**. An electron from the **M-shell** falls to fill the L-shell hole, and the resulting energy is used to eject another electron from the **M-shell**.

   (Any two differences-full marks)
2. Explain the mechanism of Internal Conversion (IC) and Intersystem Crossing (ISC). *[U] / [CO3]*

   According to the Jablonski Diagram, these are the two primary types of non-radiative transitions where an excited molecule loses energy without emitting light, Internal conversion and Inter system crossing.

   Internal Conversion (IC)

   A non-radiative transition between two electronic states of the same spin multiplicity (eg. S2 – S1). Occurs when a molecule in a high vibrational level of a lower electronic state has the same energy as the lowest vibrational level of a higher electronic state. The excess energy is lost as heat to the surroundings through molecular collisions. Extremely fast, typically occurring in 10 -14 to 10 -11 seconds.

   Intersystem Crossing (ISC)

   A non-radiative transition between two electronic states of different spin multiplicity (e.g., S1 to T1). Involves a spin flip of an excited electron. While technically forbidden by selection rules, it occurs due to spin-orbit coupling. It is the precursor to Phosphorescence. Slower than IC, typically 10 -8 to 10 -3 seconds

(Each mechanism consists of 2marks)

3. Calculate the concentration of a colloidal solution using the Beer-Lambert Law, given its absorbance, path length, and extinction coefficient. *[A] / [CO3]*

   According to the Beer-Lambert Law, The absorbance of a beam of monochromatic radiation is directly proportional to the molar absorptivity, the path length of the sample, and the concentration of the absorbing species
   Using the Beer–Lambert Law, the concentration of a colloidal solution can be calculated using the equation:
   $A = \varepsilon lc$, where:

- A = absorbance
- ε = extinction coefficient
- l = path length
- c = concentration

Rearranging the equation:

$c = A/\varepsilon l$

For Example

Given: Absorbance, A=0.50A = 0.50, , Path length, l=1 cm=1cm, Extinction coefficient, ε=100L mol−1cm−1 Hence, c= 0.50*100/1= 0.005mol/L

(Definition-2 marks; equation :1marks; explanation: 1marks)

4. Analyze the importance of combining TGA and DSC for complete thermal characterization of nanomaterials. *[An] / [CO4]*
   Thermogravimetric Analysis (TGA) and Differential Scanning Calorimetry (DSC) is essential for a complete thermal characterization of nanomaterials because each technique provides complementary information that gives a full picture of thermal behaviour.

   TGA (Thermogravimetric Analysis)
   Measures mass change as a function of temperature or time and Provides information on Thermal stability, Decomposition temperatures, Moisture and volatile content, Oxidation or reduction processes.

   DSC (Differential Scanning Calorimetry)

   Measures heat flow associated with thermal transitions and provides information on Melting point, Glass transition temperature (Tg),Crystallization behavior, Phase transitions, Enthalpy changes

(Full marks should be given if the student provides at least one comparison.)

5. How does Zeta potential depend on pH? *[U] / [CO5]*

   Zeta Potential is strongly influenced by the pH of the medium because pH affects the surface

charge of colloidal particles. As the pH changes, the concentration of hydrogen ions (H^+) and hydroxide ions (OH^-) around the particle surface also changes, altering the electrical double layer and hence the zeta potential.

- At **low pH (acidic medium)**, the particle surface tends to adsorb more H^+ ions, resulting in a **positive zeta potential**.
- At **high pH (basic medium)**, adsorption of OH^- ions increases, producing a **negative zeta potential**.
- At a particular pH called the **isoelectric point (IEP)**, the zeta potential becomes zero because the positive and negative charges balance each other.

The magnitude of zeta potential determines the stability of colloidal systems. Higher positive or negative zeta potential values indicate better colloidal stability due to stronger electrostatic repulsion between particles.

(Definition:1 marks; Explanation on effects: 3marks)

6. Write down the BET equation and identify each term involved. *[K] / [CO4]*

$$\frac{p/p_0}{n(1-p/p_0)} = \frac{1}{n_m C} + \frac{C-1}{n_m C}(p/p_0)$$

p: The equilibrium pressure of the adsorbate gas
*p*0: The saturation vapor pressure of the adsorbate at the analysis temperature
n **(or** *v***)**: The specific amount (or volume) of gas adsorbed at the equilibrium relative pressure
*nm***(or** *vm***)**: The **monolayer capacity**, representing the amount of gas required to cover the entire solid surface with a single, complete layer of molecules
C: The BET constant, which is exponentially related to the energy (heat) of monolayer adsorption

(Equation: 2 marks; Explanation of terms: 2marks)

7. Recognize the scale at which STM operates. *[K] / [CO2]*
Scanning Tunneling Microscopy (STM) primarily operates at the atomic and sub-atomic scales, providing a resolution of approximately 0.1 nanometers (1 Å) laterally (1 mark) and 0.01 nanometers (0.1 Å) vertically. (1 mark)

PART B (Short Essay)

Answer any 5 questions. Each question carries 7 marks. (5 x 7 = 35 Marks)

8. List and explain the advantages of Secondary Ion Mass Spectroscopy (SIMS). *[K] / [CO3]*
1.Analysis is fast, time-saving. Sensitive to elements heavier than helium.
2.It is extremely sensitive, 10ppm for heavy elements.
3.ISS is applicable at elevated temperature up to 1200^0C.
4.Give information at the atomic level.
5.Analyzing biological materials.
6.Qualitative identification of surface elements.
7.Identification of surface isotopes and elemental surface distribution profile.

(Minimum 7 points)

9. What is a Supporting electrolyte in Polarography? *[U] / [CO5]*
   A supporting electrolyte is an inert electrolyte added in large excess to the analyte solution to increase its electrical conductivity and eliminate the migration current. It does not participate in the electrode reaction but helps maintain a constant ionic strength throughout the experiment.

   The supporting electrolyte ensures that the movement of electroactive species toward the electrode occurs mainly by diffusion rather than by migration under the influence of an electric field. This improves the accuracy and shape of the polarographic wave. Common examples of supporting electrolytes include potassium chloride (KCl), potassium nitrate (KNO_3), and sodium sulfate (Na_2SO_4).

   (Definition :3marks; Explanation :4marks)

10. Discuss the relevance of particle size distribution with respect to the size-dependent properties of quantum dots. *[U] / [CO1]*
    To understand a sample, we have to measure hundreds of particles to create a **Histogram**.
    a) Average Particle Size (D): Calculated by measuring the diameter of at least 100–200 particles from TEM images.

    b) Standard Deviation (σ): Tells you how "uniform" your sample is.

    - **Monodisperse:** All particles are almost the same size (narrow histogram).
    - **Polydisperse:** There is a wide variety of sizes (broad histogram).[6]

    c) The Histogram: This is a bar chart where the x-axis is the **Size (nm)** and the y-axis is the **Frequency (Count)**. Usually, a "Gaussian" or "Normal Distribution" curve is fitted over the bars.

    (Give 7 marks for relevant answer)

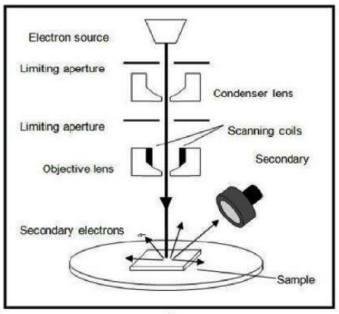
11. Explain the working of a SEM with the aid of a labeled diagram. *[K] / [CO1]*

This technique provides high-resolution imaging and detailed information about the sample's surface topography and composition. Scanning Electron Microscopy works on the principle of the interaction between a focused beam of high-energy electrons and the surface of a specimen. In SEM, a beam of electrons is generated by an electron gun and focused onto the sample surface using electromagnetic lenses. When the electron beam strikes the specimen, various signals such as secondary electrons, backscattered electrons, and characteristic X-rays are produced due to electron–matter interactions.

The emitted secondary electrons are collected by a detector and converted into electrical signals to form a highly magnified image of the sample surface. Since the electron beam scans the specimen point by point, detailed information about the surface morphology, texture, composition, and topography of the material can be obtained.

Key Principles of SEM

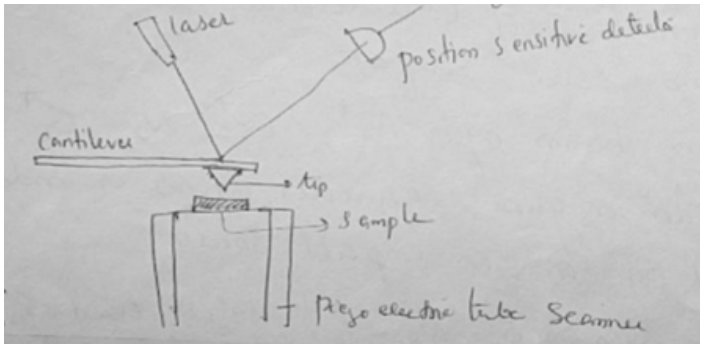
a). Electron Beam Generation and Focusing, b). Scanning and Interaction c). Signal Detection and Image Formation



(Working Principle-4 marks, Labelled diagram -3 marks)

12. Explain the working principle of AFM. *[U] / [CO2]*

Atomic Force Microscope can analyze conducting as well as non-conducting materials and provides detailed information about surface morphology, roughness, and nanoscale structures with atomic-level resolution.



It works on the principle of measuring the force of interaction between a sharp probe tip and the surface of a sample. In AFM, a very fine tip attached to a flexible cantilever scans across the surface of the specimen. As the tip moves over the sample, attractive or repulsive forces between the atoms of the tip and the sample cause the cantilever to bend or deflect.

A laser beam is focused onto the back of the cantilever, and the reflected beam falls on a photodetector. The deflection of the cantilever changes the position of the reflected laser beam, which is converted into electrical signals. These signals are processed by a computer to generate a high-resolution three-dimensional image of the surface.

(Working Principle-4 marks, Any schematic diagram -3 marks)

13. Explain the basic principle of Surface-Enhanced Raman Scattering (SERS). Discuss the two primary mechanisms responsible for the signal enhancement. *[An] / [CO3]*

Surface-Enhanced Raman Spectroscopy (SERS): Amplifies signals by factors of 10^6 to 10^{11} using localized electromagnetic fields on metallic nanostructures. SERS addresses the primary limitation of standard Raman spectroscopy—its extremely low scattering cross-section.

Mechanisms involved in SERS

(a) Electromagnetic Mechanism (EM) is the dominant contributor to signal enhancement. It occurs when the incident laser frequency matches the Localized Surface Plasmon Resonances (LSPR) of

noble metal nanostructures (like gold or silver). This creates a massive local electric field, and the resulting Raman intensity scales approximately with the fourth power of this field enhancement.

(b) A secondary effect providing a smaller boost (factors of 10^1 to 10^3), which involves charge-transfer processes between the metal surface and the adsorbed molecule, effectively acting as a resonance Raman effect.

(c) Hot Spots: These are highly localized regions, such as the nanometric gaps between adjacent particles or sharp metallic tips, where the electromagnetic field and signal enhancement are exponentially higher than on flat surfaces.

(Definition: 3 marks; Mechanisms: Each explanation contains 2 marks)

14. Discuss the principle of Small-Angle X-ray Scattering (SAXS). Explain how the "Guinier Plot" and the "Porod Law" are used. *[U] / [CO4, CO6]*

Small-Angle X-ray Scattering (SAXS) is a non-destructive technique used to quantify nanoscale density differences (typically between 0.1° and 10°) within a sample.

By analyzing the elastic scattering behavior of X-rays at very small angles, it provides structural information for dimensions ranging from 1 to 100 nm.

The Guinier plot and Porod's law represent standard mathematical approaches used in Small-Angle X-ray Scattering (SAXS) to extract the structural information described throughout the material.

1. Guinier Plot (Determining Size)

It is used to determine the averaged particle size or radius of gyration (*Rg*) of nanostructures

A Guinier analysis specifically targets the smallest recorded angles, as the sources note that larger object dimensions are probed at the smaller recorded angles

SAXS is capable of delivering structural information for dimensions between **1 and 100 nm** by recording scattering at very small angles (0.1–10°)

2. Porod's Law

It is primarily used to quantify the **surface-to-volume ratio** and provide information regarding the **porosity** and **surface area** of a material

Porod analysis relies on the **nanoscale density differences** between a particle and its matrix

(Definition of SAXS-3 marks; Explanation of laws(each with at least two points- 4 marks)

PART C (Essay)

Answer any 1 question. Each question carries 15 marks. (1 x 15 = 15 Marks)

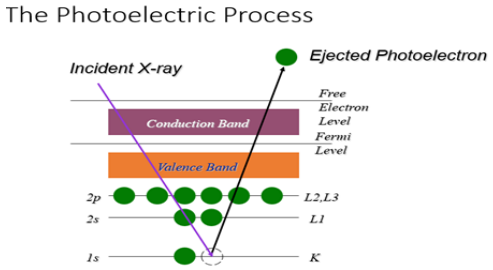
15. Discuss the working principle of X-ray photoelectron spectroscopy. Also explain how it is useful for quantitative elemental analysis. *[U] / [CO3]*

X-ray Photoelectron Spectroscopy (XPS), also known as Electron Spectroscopy for Chemical Analysis (ESCA) is a widely used technique to investigate the chemical composition of the surfaces.

Working Principle

XPS is based on the photoelectric effect. When a sample is bombarded with a beam of monochromatic X-rays, the X-ray photons interact with the inner-shell electrons of the atoms in the sample. Usually an Aluminum or Magnesium anode is used as the X-ray Source. If the energy

of the X-ray photon hv is greater than the binding energy (BE)of an electron, the electron is ejected from the atom as a photoelectron. The kinetic energy of these emitted photoelectrons is measured using an electron spectrometer.



The binding energy of the electron, which is characteristic of the element and its orbital, can be calculated by **Koopman's theorem.**

$$KE = hv - BE - \varphi$$

Where:BE= Electron Binding Energy, KE= Electron Kinetic Energy,φ= Spectrometer Work Function

Because each element has a unique set of binding energies, the resulting spectrum of intensity versus binding energy allows for the identification of all elements (except hydrogen and helium) present on the surface. The kinetic energy difference between the peak of the sample and corresponding peak in a reference compound is known as chemical shift.The shifts of the photoelectron lines in an ESCA spectrum reflect the increase in binding energy. ESCA chemical shifts lie in the range 0-1500eV.

Quantitative Elemental Analysis

XPS is highly effective for quantitative analysis because the number of photoelectrons detected for a specific element is directly proportional to its concentration within the sampled volume.To determine the relative atomic concentration of an element, the following factors are used.

1. Peak Area
2. Relative Sensitivity Factors (RSF)
3. Calculation of Atomic Fraction. Refer to any example.

(Working principle with figure and equation : 8 marks ,Quantitative elemental analysis with example: 7 marks)

16. Describe the basic instrumental features of FESEM with a schematic diagram. *[U] / [CO1]*

The fundamental principle of **FESEM (Field Emission Scanning Electron Microscopy)** lies in its unique method of generating electrons. While a standard SEM uses heat to "boil" electrons off a filament, FESEM uses a powerful electric field to pull them from a sharp needle-like tip. This results in a beam that is much thinner, more coherent, and significantly higher in brightness, allowing for ultra-high-resolution imaging down to **1 nanometer**.

1. The Core Principle: Field Emission

The heart of the FESEM is the **Field Emission Gun (FEG)**.

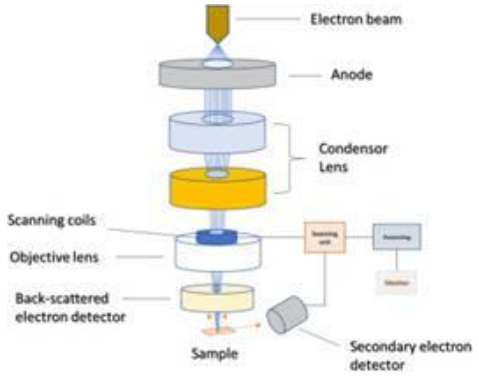
The Emitter: The source is usually a single-crystal tungsten wire with an extremely sharp tip (radius < 100 nm).

Quantum Tunneling: Instead of using heat (thermionic emission), a high voltage (extracting voltage) is applied to the tip. This creates a massive electric field gradient that allows electrons to "tunnel" through the energy barrier of the metal and escape into the vacuum.

2. Working Mechanism

Once the electrons are "pulled" from the tip, the process follows these steps:

a. Acceleration
b. Focusing
c. **Raster Scanning**
d. **Signal Generation**
e. **Detection & Image Construction**



(Principle -4 marks, working mechanism-7 marks, Schematic diagram- 4 marks)